MSG No. 21.43 C_a222 [Type IV, orthorhombic]

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det	TR
1	$\{1\|0\}$	$\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\end{bmatrix}$	1	1
2	$\{2_{100}\|0\}$	$\begin{bmatrix}1&0&0&0\\0&-1&0&0\\0&0&-1&0\end{bmatrix}$	1	1
3	$\{2_{010}\|0\}$	$\begin{bmatrix}-1&0&0&0\\0&1&0&0\\0&0&-1&0\end{bmatrix}$	1	1
4	$\{2_{001}\|0\}$	$\begin{bmatrix}-1&0&0&0\\0&-1&0&0\\0&0&1&0\end{bmatrix}$	1	1
5	$\{1\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}1&0&0&\frac{1}{2}\\0&1&0&\frac{1}{2}\\0&0&1&0\end{bmatrix}$	1	1
6	$\{2_{100}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}1&0&0&\frac{1}{2}\\0&-1&0&\frac{1}{2}\\0&0&-1&0\end{bmatrix}$	1	1
7	$\{2_{010}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&1&0&\frac{1}{2}\\0&0&-1&0\end{bmatrix}$	1	1
8	$\{2_{001}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&-1&0&\frac{1}{2}\\0&0&1&0\end{bmatrix}$	1	1
9	$\{1'\|\frac{1}{2}00\}$	$\begin{bmatrix}1&0&0&\frac{1}{2}\\0&1&0&0\\0&0&1&0\end{bmatrix}$	1	−1
10	$\{2_{100}'\|\frac{1}{2}00\}$	$\begin{bmatrix}1&0&0&\frac{1}{2}\\0&-1&0&0\\0&0&-1&0\end{bmatrix}$	1	−1
11	$\{2_{010}'\|\frac{1}{2}00\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&1&0&0\\0&0&-1&0\end{bmatrix}$	1	−1
12	$\{2_{001}'\|\frac{1}{2}00\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&-1&0&0\\0&0&1&0\end{bmatrix}$	1	−1
13	$\{1'\|0\frac{1}{2}0\}$	$\begin{bmatrix}1&0&0&0\\0&1&0&\frac{1}{2}\\0&0&1&0\end{bmatrix}$	1	−1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
14	$\{2_{100}'\|0\frac{1}{2}0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	-1
15	$\{2_{010}'\|0\frac{1}{2}0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	-1
16	$\{2_{001}'\|0\frac{1}{2}0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	-1